

20 GeV halo-like excess of the Galactic diffuse emission and implications for dark matter annihilation (1)

具体的な検出方法と、差し引き方法

使用データ

Fermi-LAT public data

- Pass 8 / UltraClean class
- All-sky survey data
- 期間：2008–2023（15年分）

2.1 The LAT data and the region of interest

This work is based on the Pass 8 UltraClean class of the 15-year public *Fermi* LAT data from 2008 August 4 to 2023 August 2. In addition, a zenith angle cut of $\theta < 100^\circ$ is adopted

<https://fermi.gsfc.nasa.gov/cgi-bin/ssc/LAT/LATDataQuery.cgi>

The screenshot shows the 'LAT Photon, Event, and Spacecraft Data Query' web interface. It features a sidebar on the left with a 'Catalog' and 'Library' section. The main form includes the following fields and controls:

- Object name or coordinates:** A text input field.
- Coordinate system:** A dropdown menu set to 'J2000'.
- Search radius (degrees):** A text input field.
- Observation dates:** A text input field.
- Time system:** A dropdown menu set to 'Gregorian'.
- Energy range (MeV):** A text input field.
- LAT data type:** A dropdown menu set to 'Photon'.
- Spacecraft data:** A checkbox that is checked.
- Buttons:** 'Start Search' and 'Reset' buttons at the bottom.

The region of interest (ROI) in this study is set to $|l| \leq 60^\circ$ and $|b| \leq 60^\circ$ around the GC, and the disk area of $|b| < 10^\circ$ is excluded from the dark matter search. Photon count maps and exposure maps with pixel scales of $\Delta l_p = \Delta b_p = 0.125^\circ$ in Cartesian coordinates are created from the public *Fermi* LAT data and tools², resulting in 960×960 pixels in the ROI including the disk region.

まず
地球大気 (Earth limb) を減らすため zenith angle cut : $\theta < 100^\circ$ を入れる
1 GeV以上だけ使う

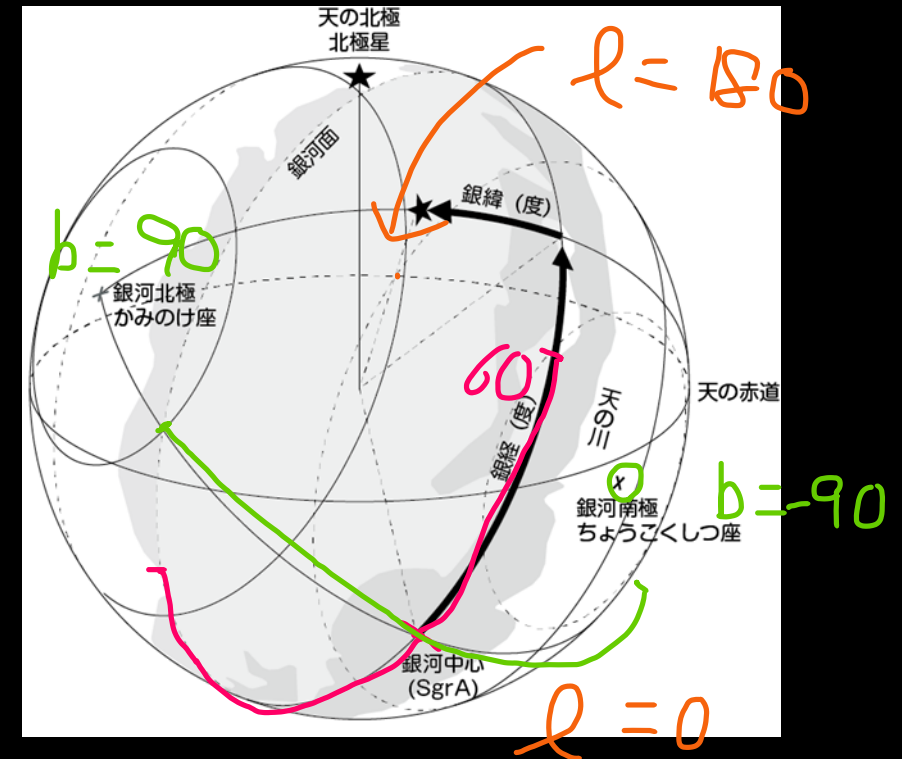
エネルギーを13個のビンに分ける

- 13エネルギービン (対数等間隔) に分割
- ビン中心の最小・最大は 1.51 GeV と 814 GeV

空の範囲 (ROI) を切り出す

- ROI : $|l| \leq 60^\circ$, $|b| \leq 60^\circ$
- ただし $|b| < 10^\circ$

photon count map と exposure map を作成



2.2 Likelihood analyses = 尤度 (ゆうど) =

「このモデルが本当だとしたとき、
今見えているデータが出る“もっともらしさ L ”」 by チャッピー

① 尤度が減らないから解像度を下げた ($0.125^\circ \Rightarrow 10^\circ$)

②
尤度を“最大化”したい！

いまあるものを整理すると：

観測 $\Rightarrow C(\text{obs}, i)$ (10° セルごとのフォトン数)

モデル $\Rightarrow C(\text{exp}, i) =$ (テンプレート T_l を係数 f_l で足したもの)

expectations are compared in each pixel. Since the halo-like component is expected to be much smoother than the pixel scale of 0.125° , calculating the likelihood at a coarser resolution allows computation time to be reduced without compromising dark matter search. Therefore, we

Let $C_{\text{obs}, i}$ be the observed photon count in the i -th cell and $C_{\text{exp}, i}$ be the number of photons expected from the model, and then the likelihood function L based on Poisson statistics is

$$\ln L = \sum_i [C_{\text{obs}, i} \ln(C_{\text{exp}, i}) - C_{\text{exp}, i}] , \quad (2.1)$$

where the sum is for all cells. The expected count $C_{\text{exp}, i}$ is calculated by summing up the expected counts over all pixels in the cell:

$$C_{\text{exp}, i} = \sum_k F_{m, \text{tot}}(\Omega_k) A_k \Delta\Omega_k \Delta E , \quad (2.2)$$

where k represents each pixel, $F_{m, \text{tot}}$ is the total model diffuse photon flux per unit photon energy and solid angle to the angular direction Ω_k of the k -th pixel, A_k and $\Delta\Omega_k$ are the LAT exposure (in units of $\text{cm}^2 \text{s}$) and the solid angle of the k -th pixel, and ΔE is the width of the photon energy bin under consideration. Though the expected counts should be convolved

The model diffuse flux $F_{m, \text{tot}}$ is a linear combination of various map templates:

$$F_{m, \text{tot}}(\Omega_k) = \sum_l f_l F_l(\Omega_k) , \quad (2.3)$$

where $F_l(\Omega_k)$ is the template map of the l -th model component. Then the best-fit values and statistical errors of the fitting parameters f_l are obtained by maximizing the likelihood function using the Markov chain Monte Carlo (MCMC) method.

テンプレート T_l を係数 f_l で足したもの??

要素↑ と ↑強さのつまみ 【by チャッピー】

テンプレート

- (1) 点源、
- (2) 宇宙線相互作用による拡散的星間放射、
- (3) 等方放射、
- (4) ループI、
- (5) フェルミ・バブル、
- (6) ハロー状放射

これらが、考えられるガンマ線源

テンプレートの説明！



2.3 Model map templates

Following the analyses on the Fermi bubbles and the GC GeV excess by the *Fermi* LAT team [7, 39], the following components are considered to fit the gamma-ray maps in the ROI: (1) point sources, (2) diffuse interstellar emission by cosmic-ray interactions, (3) isotropic radiation, (4) Loop I, (5) the Fermi bubbles, and (6) a halo-like emission. The modeling of these components will be explained in order below. However, since the Fermi bubble template will be created during the fitting process in this study, it will be described in section 3.1.

The template of point sources in the LAT 14-year Source Catalog³ (4FGL-DR4, `gll_psc_v35.fit`) [42, 43] is calculated using the parametrized spectra from the catalog and the energy-dependent LAT point spread function (PSF). 4FGL-DR4 also provides a catalog of extended sources, in addition to the point sources. Since the extended sources are not many, circular regions with a radius twice the semi-major axis of the catalog are masked and excluded from the analysis.

Gamma rays produced by cosmic rays propagating in the MW are the most significant component of the Galactic diffuse gamma-ray emission. Here, we utilize the GALPROP code⁴ (ver. 54) [44–46], which numerically solves the generation and propagation of cosmic rays and calculates the gamma-ray intensity resulting from interactions with interstellar gas and radiation fields. Among the various GALPROP parameter sets used in the LAT team’s study of the Galactic diffuse emission [47], $S_L Z_6 R_{30} T_{150} C_2$ is adopted as the baseline model in the present work. This parameter set is used for the inverse Compton scattering (ICS) component in the Galactic interstellar emission model (GIEM, `gll_iem_v07.fits`)⁵ [48, 49], which is recommended by the LAT team for normal point source analysis. The GALPROP model radiation is divided into two components, gas and ICS, which are fitted independently. The gas component is the sum of pion-decay gamma rays produced by cosmic-ray hadrons, and bremsstrahlung from cosmic-ray electrons and positrons. The flux ratio between pion decay and bremsstrahlung is fixed to the original model value, making the gas component a single template. The GALPROP model emission can be further divided into different components, and the fitting results with a larger number of sub-components or different model parameters will be discussed in section 3.4.4.

Loop I is a giant structure spanning more than 100° in the radio sky [50], and although it is unclear whether it is nearby supernova remnants or an outflow from the GC [51, 52], it has also been detected by LAT [53]. We adopt the geometric model of Loop I, which assumes a uniform emissivity within two shells with radii of 50–100 pc, based on the nearby hypothesis.

The parameters of the geometric model are taken from the LAT team papers [7, 39], which differ slightly from the values in Wollben’s original paper [51]. The normalization factors for each of the two shells are treated as two independent fit parameters (f_l).

To calculate the halo component model map, we first assume that the density profile of the MW dark matter halo is NFW, and use the parameters from the *Via Lactea II* simulation [54, 55]. Therefore, the density profile within the virial radius $r_{\text{vir}} = 402$ kpc is $\rho(r) = \rho_s x^{-1}(1+x)^{-2}$, where $x = r/r_s$, $r_s = 21$ kpc, and $\rho_s = 8.1 \times 10^6 M_\odot \text{kpc}^{-3}$. The distance from the GC to the solar system is set at 8 kpc, and hence $\rho_\odot = 0.42 \text{ GeV cm}^{-3}$. Annihilation gamma-ray diffuse flux is proportional to the line-of-sight integration $\int \epsilon_\gamma dl$ of the gamma-ray emissivity ϵ_γ per unit volume. If dark matter is distributed smoothly, the emissivity obeys $\epsilon_\gamma \propto \rho^2$, and the map template in this case is denoted as NFW- ρ^2 . According to numerical simulations, fine substructures and subhalos exist within halos composed of cold dark matter, and subhalos with high internal densities may dominate annihilation gamma-ray luminosity. If the number density of subhalos is proportional to the smooth density profile, the gamma-ray emissivity would be $\epsilon_\gamma \propto \rho^1$. This emissivity also applies when dark matter particles decay rather than annihilate. Therefore, the map template of $\epsilon_\gamma \propto \rho^1$ (NFW- ρ^1) will also be tested. Subhalos may be destroyed by strong tidal forces near the center of the parent halo [54], in which case emissivity in the central region would be suppressed. While a model for this case is not tested in this work, this possibility should be kept in mind. Previous studies on the GC GeV excess found that the density profile of $\rho \propto r^{-1.25}$ ($\epsilon_\gamma \propto r^{-2.5}$) is preferred, and hence the template of $\epsilon_\gamma \propto \rho^{2.5}$ with the original NFW (NFW- $\rho^{2.5}$) is adopted as the model of the GC excess, whose emissivity becomes $\epsilon_\gamma \propto r^{-2.5}$ in the central region ($\rho \propto r^{-1}$).

The map morphologies of the point sources and GALPROP have photon energy dependencies determined by the models, while other model maps do not depend on photon energy. The initial values of f_l in the MCMC chain are the same as the original normalization for the models of point sources and GALPROP. The initial f_l of the isotropic component is set to $E^2 dN/dE = 10^{-4} \text{ MeV cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$, which is typical of the reported fluxes of the unresolved extragalactic gamma-ray background radiation [56]. For other components, the initial values of the MCMC chain are set to zero. During the MCMC fitting, the fitting parameters are limited to the range $f_l > 0$ for known components such as point sources, the GALPROP models, the isotropic component, and Loop I, but $f_l < 0$ is allowed for the halo components, which are explored as unknown components.

ここまででやったこと

Fermi-LAT のフォトンデータとエクスポージャーデータを用意して、
l と b がそれぞれ ± 60 度で、銀河面の強い放射を避けるために $|b| < 10$ 度は最初から除外。
エネルギーは 1 GeV 以上、対数的に区切った 13 個のエネルギービンごとに独立に解析する。

その上で、ガンマ線の原因候補ごとに「形が決まった地図」を用意。

点源、宇宙線と星間物質の相互作用による拡散放射、等方放射、Loop l、フェルミ・バブル、そしてハロー状放射のテンプレートを作る。これらの地図の形は固定で、変えるのはそれぞれがどれくらい強いのかという係数だけ。

各セルについて、これらのテンプレートを係数付きで足し合わせたモデルフラックスを作り、それにエクスポージャー、立体角、エネルギービン幅を掛けて足し合わせることで、そのセルで理論的に期待されるフォトン数 C_{exp} を計算する。

そして観測値 C_{obs} と予測値 C_{exp} を使って、ポアソン統計に基づく対数尤度 $\ln L$ を計算する。

$$\ln L = \sum_i [C_{\text{obs},i} \ln(C_{\text{exp},i}) - C_{\text{exp},i}] ,$$

マルコフ連鎖モンテカルロの略
一言で言うと、
「尤度が大きいパラメータの場所を、
確率的にうろうろ歩いて見つける方法」
By チャッピー

最後に、各成分の強さの係数を動かしながら $\ln L$ が最大になる点を MCMC で探す。

つまり「この成分をどれくらい入れると、観測データを一番よく説明できるか」を定量的に決めるところまでが、今ここでやったこと。

L （もっともらしさ）が最大になる f を探するのが目的

過去の研究から妥当そうなパラメーターを入れて変
化させていて

L が前回のより大きければその f に行くみたいな操作

この時小さいけど近い値を捨てずに
確率的に採用するか

決めるというのが

マルコフ連鎖モンテカルロ by
チャッピー

それをするとなにがいいの？

それをするとなんがいいの？

別に1個Lが大きく出るfが分かれば
良くない？

それをするとなんがいいの？
別に1個Lが大きく出るfが分かれば
良くない？

結論から言うと、
最大のfを1個だけ知っても「どれくらい信じて
いいか」が分からないから不十分。

理由を順に言う。

まず、logL が最大になる f が1個分かったとして
も、それが
「ガチでそこしかダメなのか」
「そのへん一帯ならどれでもほぼ同じなのか」
が分からない。

たとえば最大値が logL=100 だったとして、
そのすぐ近くに 99.9 や 99.8 が山ほどある場合
と、
100 の一点だけが鋭く突き出ている、周りが 90
とかに落ちる場合では、
意味がまったく違う。

前者なら
「f はだいたいこのへんだけど、多少ズレても同じ
くらい説明できる」
後者なら
「f はほぼこの値に固定されている」
という結論になる。

これはつまり
誤差（不確かさ）が分かるかどうかの違い。

ガス、ICS、バブル、ハローみたいな成分が同時
にあると、
ハローを少し強くすると、ICS を少し弱くしても
同じくらい合う、
みたいな状況が普通に起きる。



最大値1点だけ見ても、
「ハローが強いからなのか」
「他の成分で吸収できてるのか」
が分からない。

MCMC で周辺を取ると、
「この方向には自由度があるけど、この方向には
ない」
というパラメータ間の相関が見える。

これが見えないと、
「ハローが必要です！」
と言っても
「いや、他の成分の不確かさじゃない？」
って突っ込まれたら終わる。

この論文で本当にやりたいのは
「最適な f を決めること」じゃなくて、
「ハロー成分を入れると、どれだけ統計的に良くなるか」。

つまり
 $\ln L_{\text{max}}(\text{ハローあり}) - \ln L_{\text{max}}(\text{ハローなし})$
を信じていいかどうか。

その差が
「偶然の揺らぎじゃない」
「モデルの不確かさに埋もれてない」
と示すには、
山の形を知っておく必要がある。

だから答えはこう。

1個だけ最大の f を知るのとは「当てはめ」。
周辺も含めて見るのが「検証」。

この論文は後者をやってる。
だから MCMC を使って、99.9 や 99.8 を捨てない。

フェルミバブルはさっきのテンプレートの手法
使えないからバブルの形を作りなおそう

1.5GeVと4.3GeVのデータを使い、 $|b| < 10$ も含めて考えた。

「今の仮バブルテンプレート」と「フィットした後に
残った残差」を足し合わせることをして、より現実に近いバブル画像を作る。

それをフォトン数の地図から、エクスポージャを使って
物理量としてのフラックスの地図に直す。さらに、ノイズを抑えて大きな構造だけを見るために、 $\sigma = 1$ 度のガウシアンで平滑化する。

その平滑化された画像を見ながら、

「ここまでは明らかにバブル」

「ここから先は違う」

という判断をして、フラット・バブルテンプレートの境

3 Search for a halo-like excess

3.1 Construction of the Fermi bubble template

Unlike Loop I and the MW halo components, there are no parameterized analytical models for the Fermi bubbles. Therefore, the map template for the bubbles used in this study is made here, mostly following the methods adopted in the LAT team's papers [7, 39]. Two photon energy bins, 1.5 and 4.3 GeV, are selected for this procedure, which are in the typical photon energy range where the Fermi bubbles are prominent. As the first step, a flat (i.e., isotropic) template with sharp edges within a rough area of the Fermi bubbles is assumed, and it is fit to the data together with the other model components. Here, the GC GeV excess is included in the fit as a known component, but other halo components are not considered. At this stage, the fit is performed in the whole ROI, including the Galactic disk region ($|b| < 10^\circ$), to ensure an accurate fit to the GC excess.

Figure 1. The image of the Fermi bubbles (the flat bubble template plus fit residuals, after Gaussian smoothing with $\sigma = 1^\circ$) in the two photon energy bins (1.5 and 4.3 GeV). The boundaries of the final flat template are indicated by gray lines. The gray circular regions are excluded from the analysis due to the catalogued extended sources. The large regions at $(l, b) \sim (-50^\circ, 20^\circ)$ and $(-60^\circ, -45^\circ)$ are Cen A lobes and the SMC, respectively.

The best-fit image of the bubble is created from the sum of the flat template and fit residuals; the photon count map of this is converted into physical flux at each pixel using the exposure data, and then smoothed with a Gaussian filter with $\sigma = 1^\circ$. The energy-independent boundary of the flat template is iteratively improved by looking at these best-fit images. The image of the Fermi bubbles obtained in this way in the two energy bins is shown in figure 1, together with the final boundary of the flat template. Since the template of the Fermi bubbles will only be used at $|b| \geq 10^\circ$ in subsequent halo component searches, the Galactic disk region ($|b| < 10^\circ$) is not shown. Residuals of bright point sources are visible in some places, and these are likely due to uncertainties in the catalog spectral models and PSF. The fit in this study is performed on a cell-by-cell basis with the cell size of $\Delta l_c = \Delta b_c = 10^\circ$, and PSF errors should have little effect on the results.

The best-fit spectra of all the model components by the fit with the final flat bubble template are presented in figure 2. Here, the mean background flux within the ROI, including the Galactic disk region, is shown. For comparison with later analysis, figure 3 shows the flux excluding the disk, though the model fitting is the same as figure 2, i.e., including the disk. The spectra of the model templates are generally consistent with those of similar analyses by the LAT team [7], indicating that there are no serious problems or errors in this analysis. In particular, the peak photon energy and the flux of the GC GeV excess ($2.5 \times 10^{-9} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1} \text{ MeV}^{-1}$ at 1° from the GC at 2.5 GeV) are also consistent with those reported in the literature [8].

The morphology of the Fermi bubbles is also consistent with what has been reported in

fermiの光子数データを
exposure(その方向をどのくらいの時間どのくらいの感度
で観測したか)で割って、
条件一致させて、
物理量に直すためにフラックスにする。

次に、巨体構造だから細かなくていいから
pixel ($\sigma=1^\circ$) にした。

観測データ (Fermiが実際に見た空の明るさ) — モデル (さっきの)

この差を人間が目で見える。

ここで見るポイントはこれだけ。

この差が、

- ・ランダムなノイズっぽいか？
- ・それとも、広い範囲にわたって同じ形で残ってるか？⇒
構造があったらここはバブルだろうとする。

(ほかの文献と比べても整合性あるよ)

Figure 1. The image of the Fermi bubbles (the flat bubble template plus fit residuals, after Gaussian smoothing with $\sigma = 1^\circ$) in the two photon energy bins (1.5 and 4.3 GeV). The boundaries of the final flat template are indicated by gray lines. The gray circular regions are excluded from the analysis due to the catalogued extended sources. The large regions at $(l, b) \sim (-50^\circ, 20^\circ)$ and $(-60^\circ, -45^\circ)$ are Cen A lobes and the SMC, respectively.

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The morphology of the Fermi bubbles is also consistent with what has been reported in

見えてきたフェルミバブルの構造テンプレート

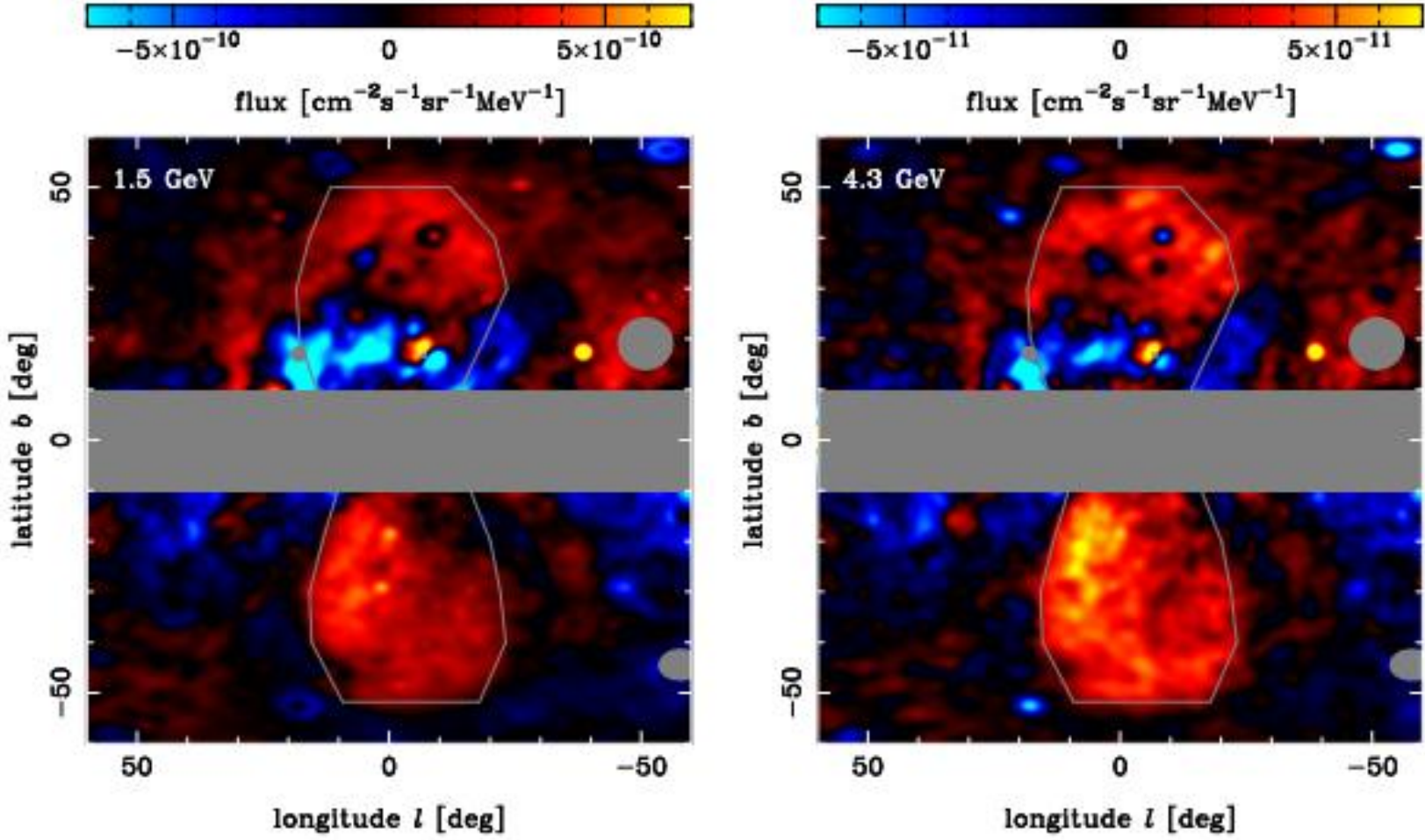


Figure2 フェルミ・バブルあり + 銀河円盤 ($|b| < 10^\circ$) を含む
= ROI 全体の平均スペクトル

Figure 3 フェルミ・バブルあり + 銀河円盤を除外 ($|b| < 10^\circ$ をマスク)
= 円盤の影響を除いたスペクトル

ほぼ同じ

横軸 = ガンマ線のエネルギー
縦軸 = 明るさ (フラックス)

- ・ 青 = フェルミ・バブル
- ・ 黒 = ガス
- ・ 緑 = 点源
- ・ 赤 = 銀河中心過剰 (GCE)

円盤を消しても
フェルミ・バブルが同じなら
「円盤の見間違いじゃない」
と言える。

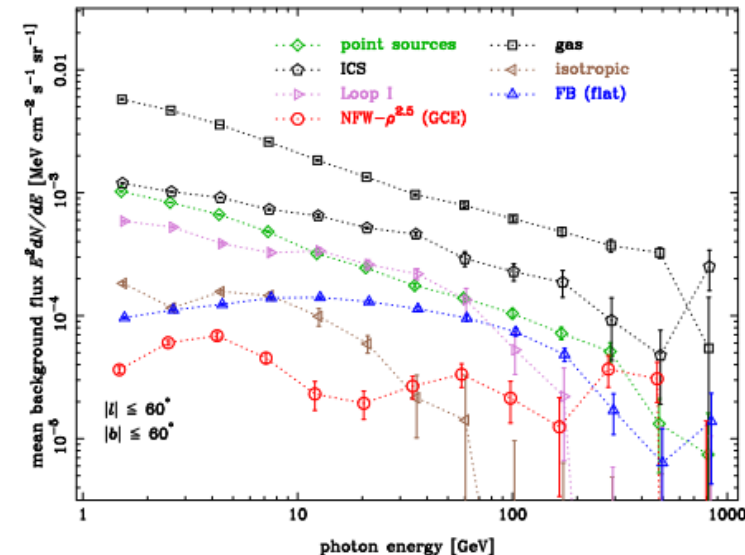


Figure 2. The best-fit spectra of model map templates, for the fit including the flat template of the Fermi bubbles (FB) and the GC GeV excess (GCE). The error bars are statistical 1σ estimated by MCMC. The fit is performed including the Galactic disk region ($|b| < 10^\circ$), and the mean background flux within the ROI, including the disk, is shown. To prevent symbols and error bars from overlapping, some symbols are slightly shifted to the left or right of the true energy bin center.

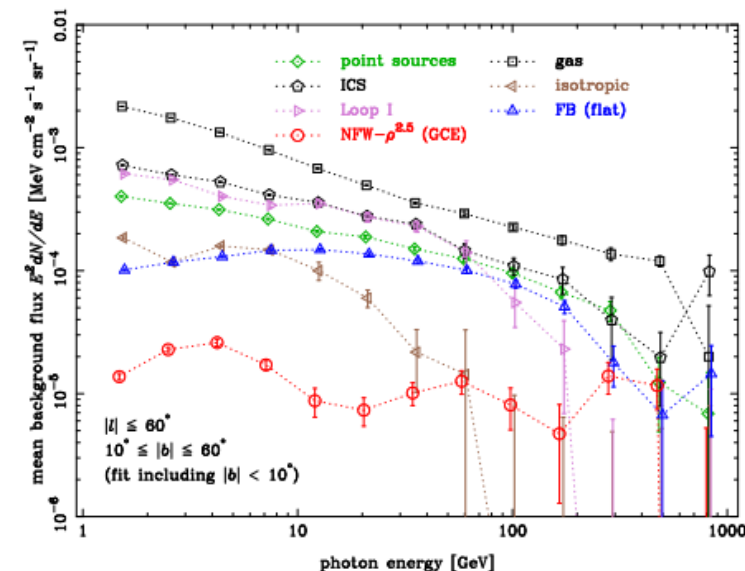


Figure 3. The same as figure 2, but showing the flux excluding the Galactic disk region ($|b| < 10^\circ$), while the fitting is the same as figure 2 (including the disk).

